CSE 4621

Machine Learning

Topic: Regularization

Source: DeepLearning.AI

Presented by

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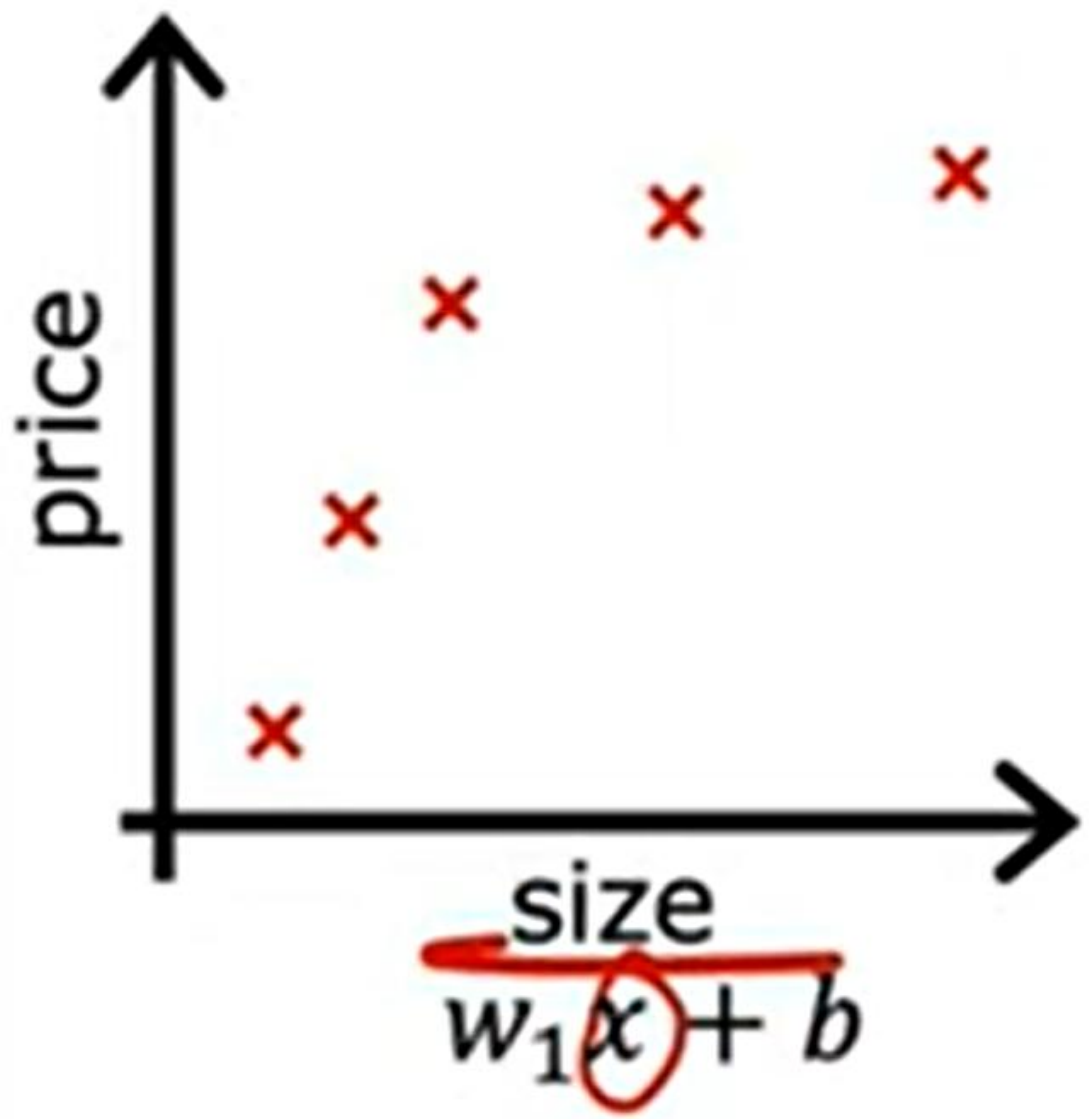
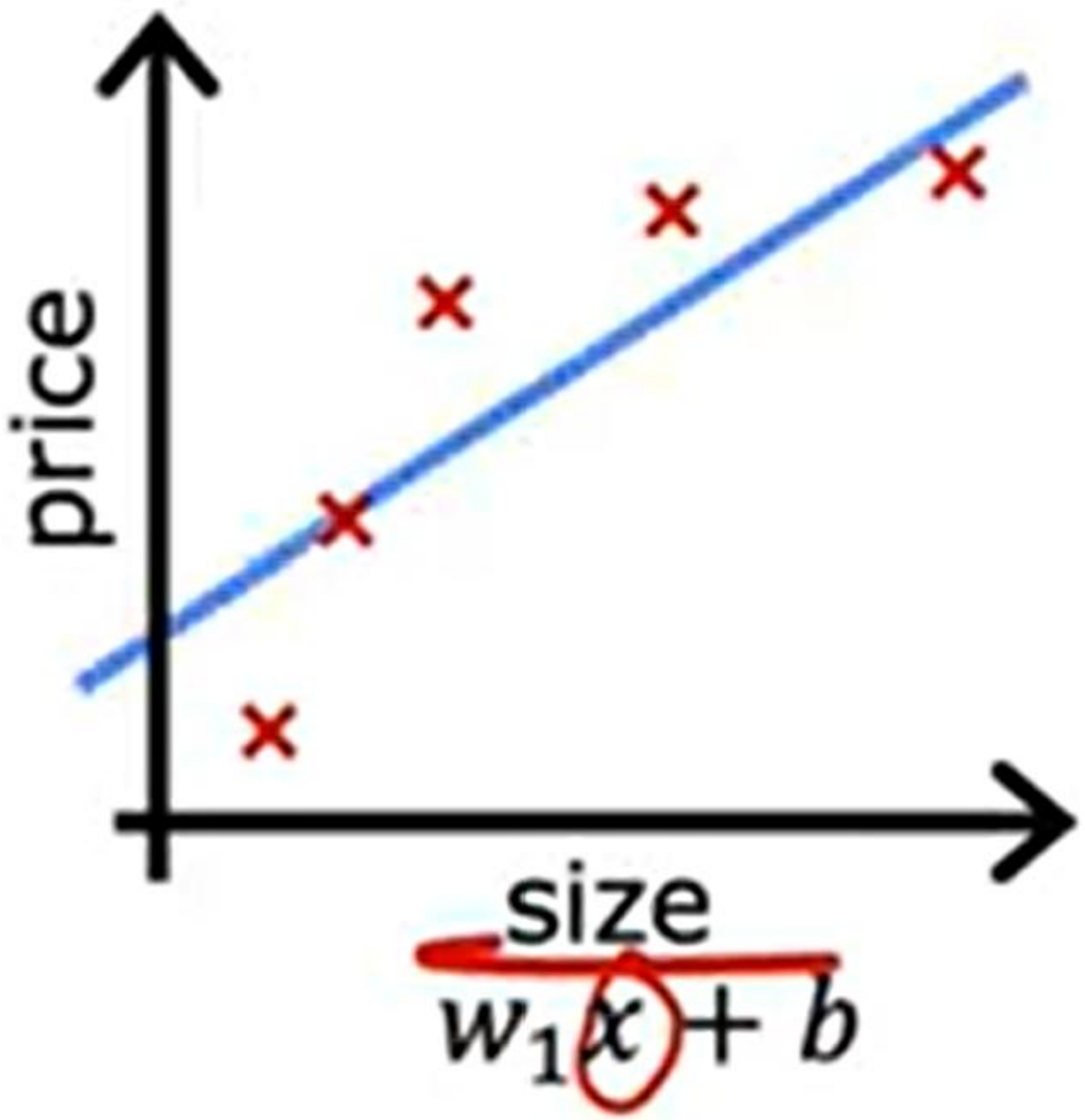
Assistant Professor

Regression example



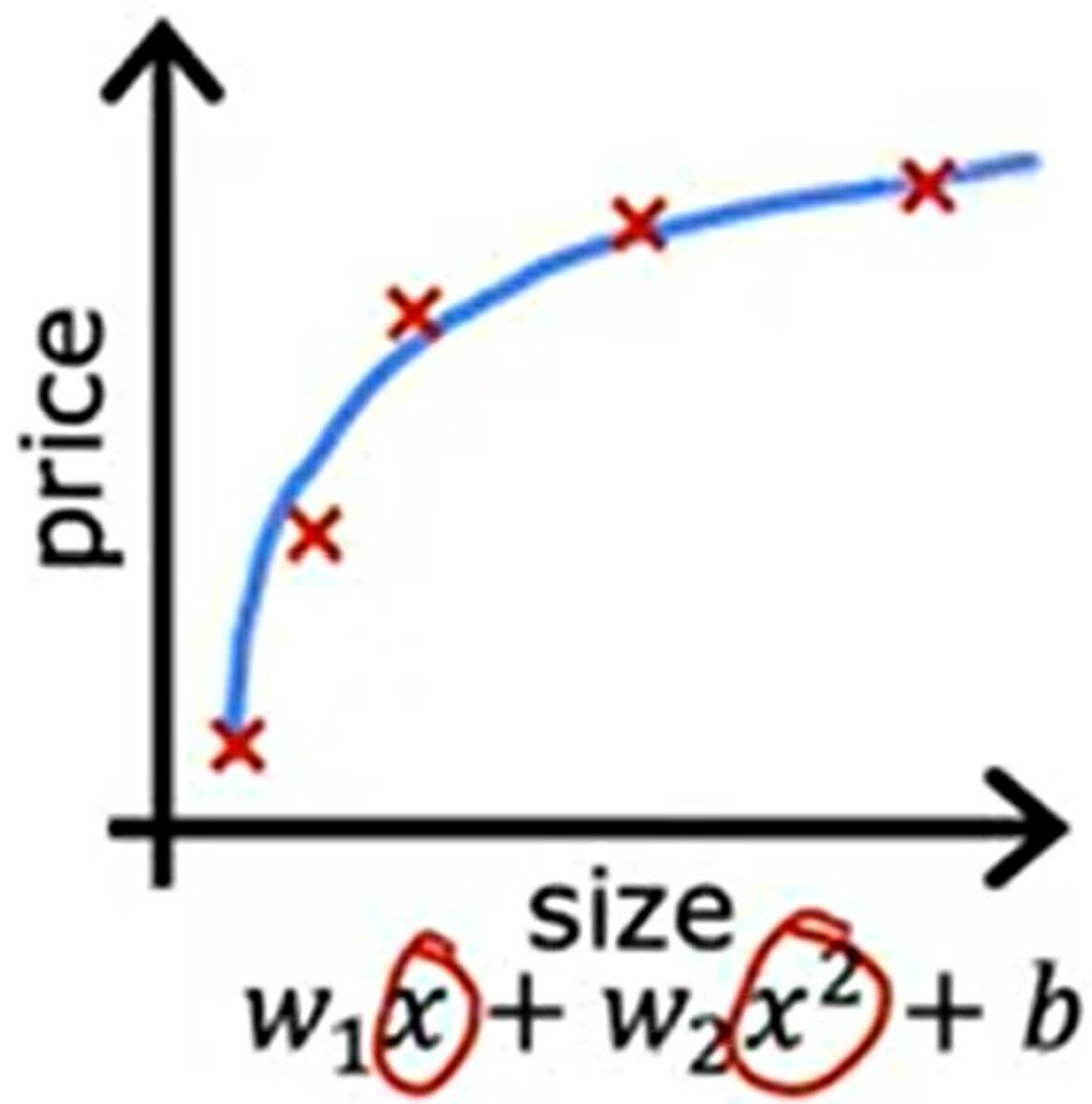


underfit

- Does not fit the training set well

high bias

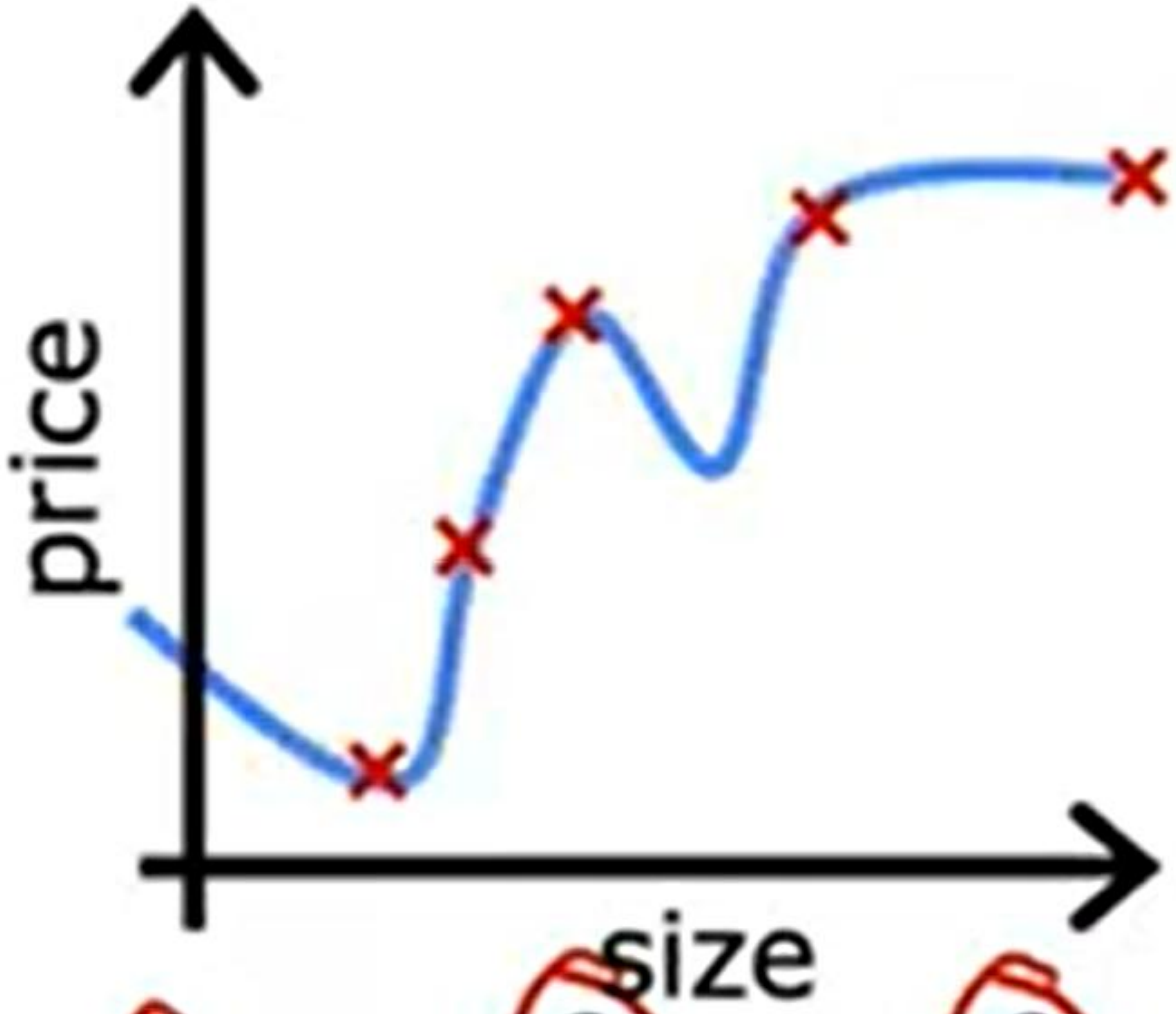
Regression example



- Fits training set pretty well

generalization

Regression example



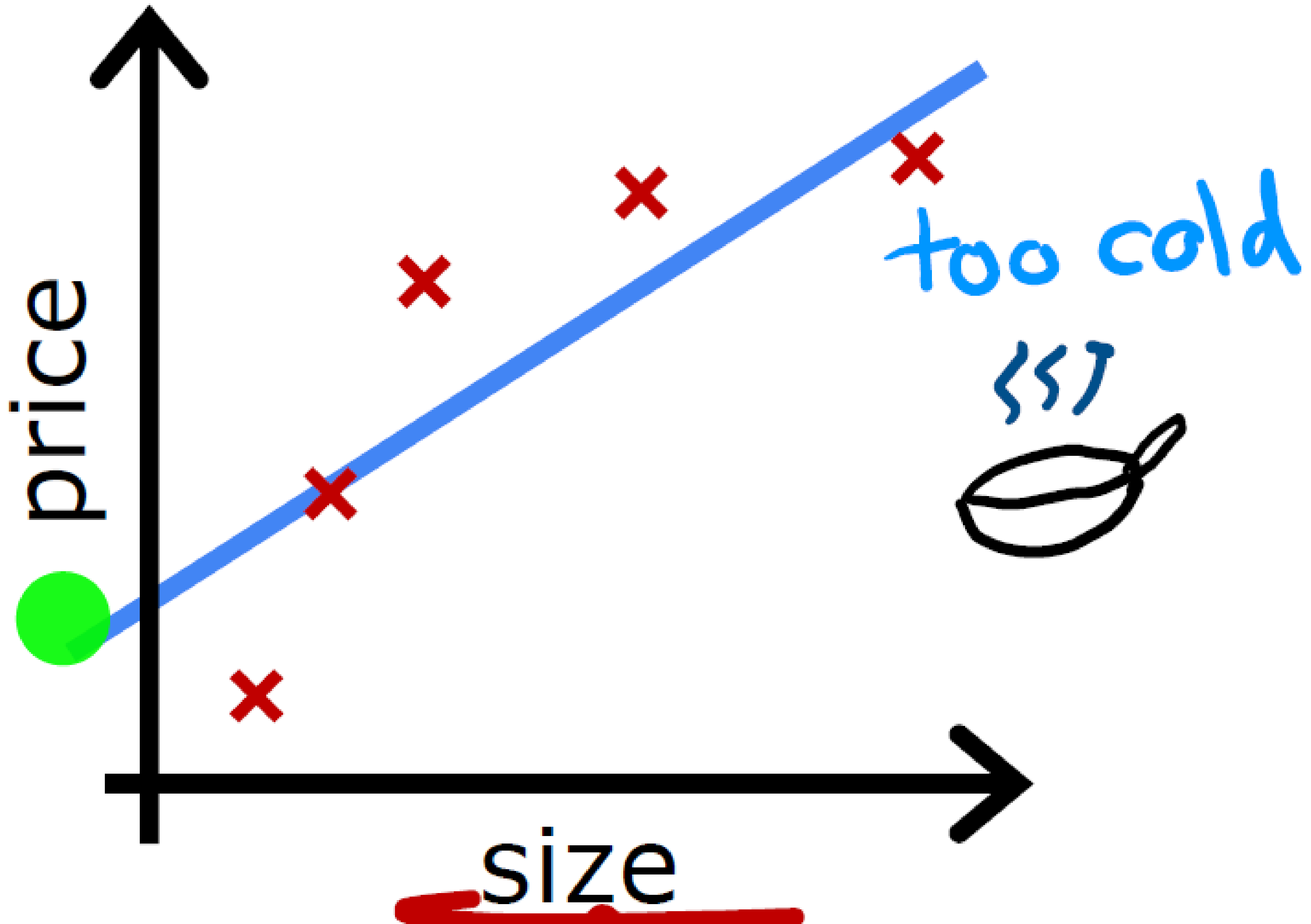
$$w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

overfit

- Fits the training set extremely well

high variance

Regression example

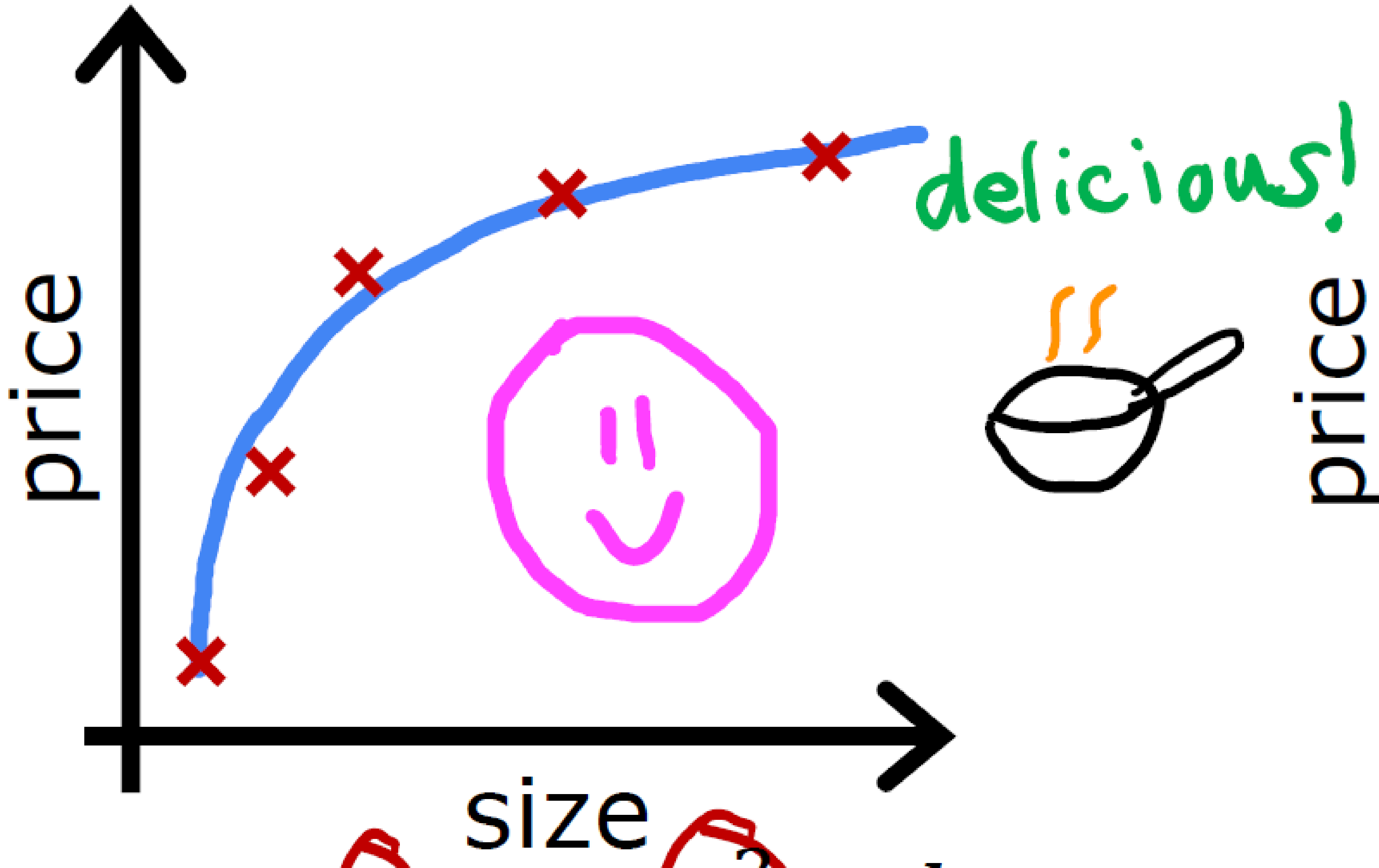


$$w_1 x + b$$

underfit

- Does not fit the training set well

high bias

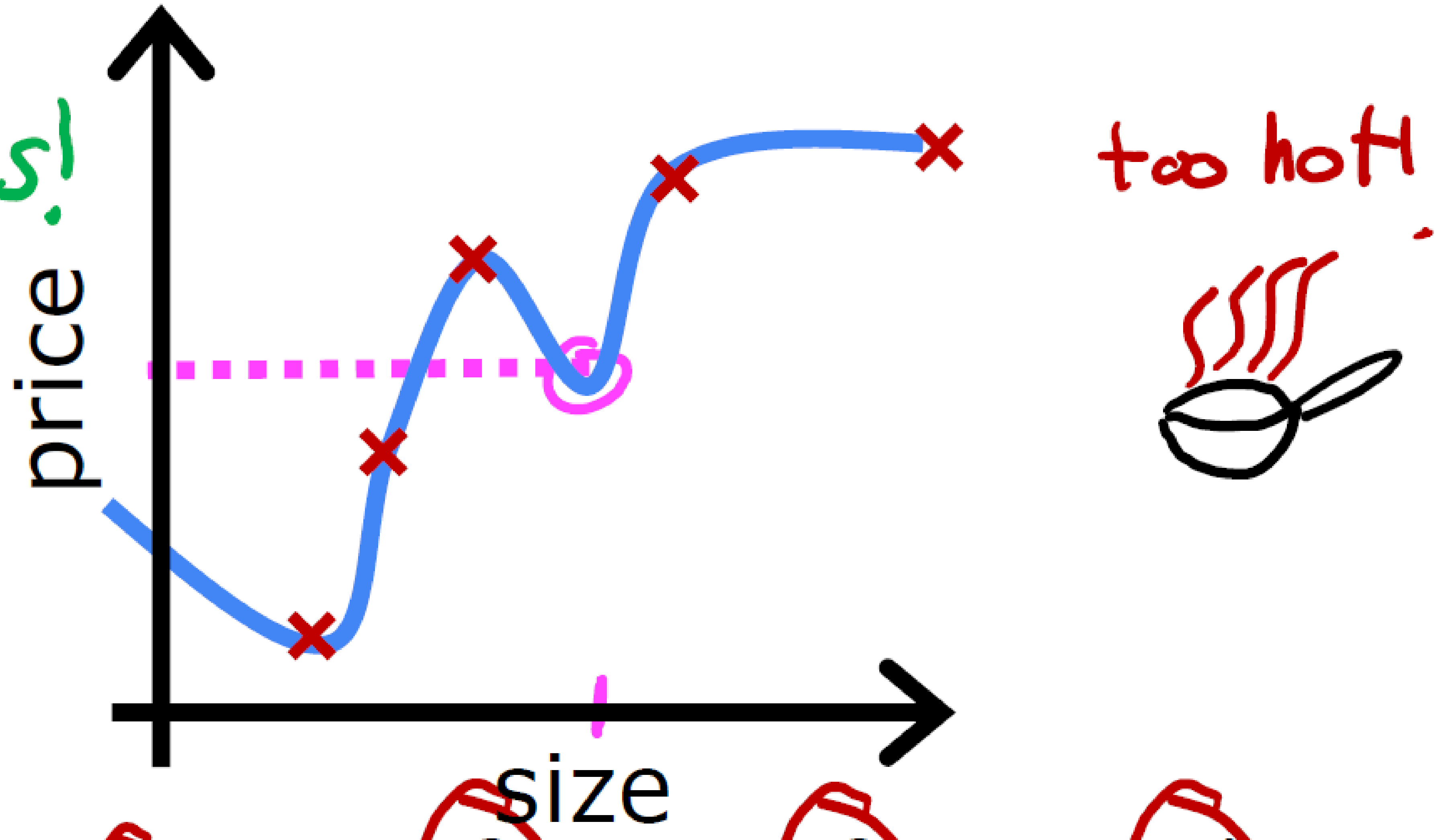


$$w_1 x + w_2 x^2 + b$$

just right

- Fits training set pretty well

generalization



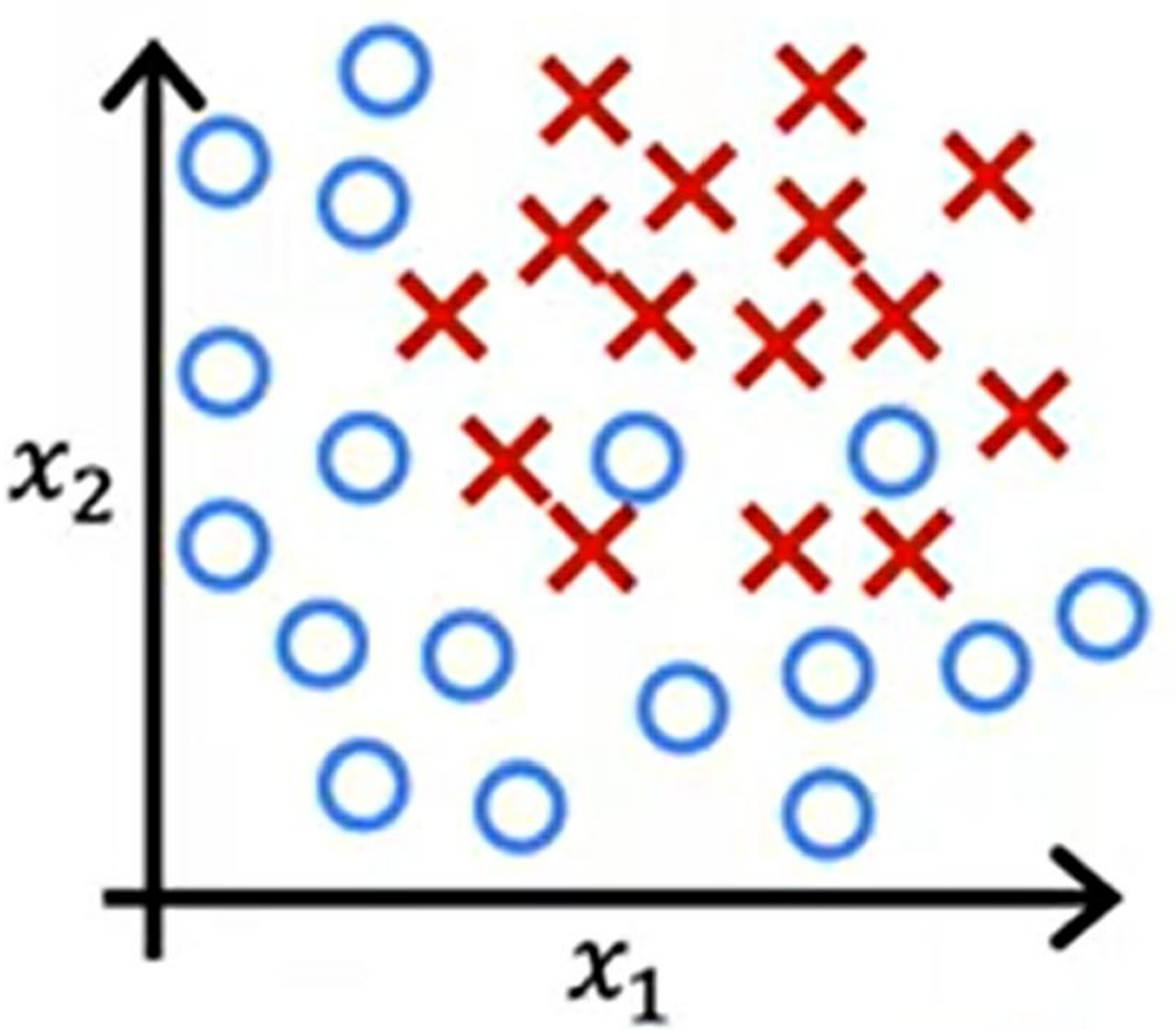
$$w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

overfit

- Fits the training set extremely well

high variance

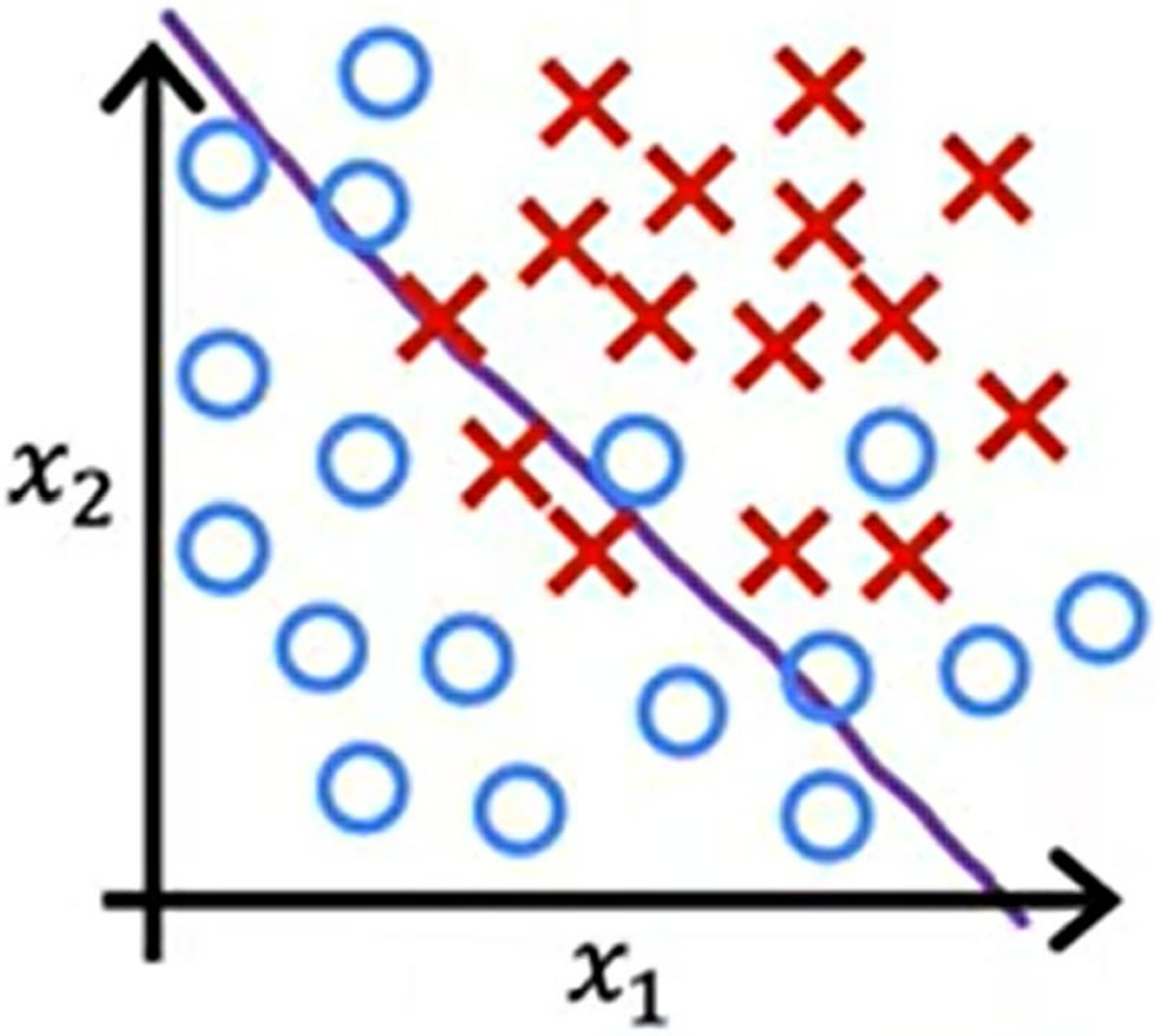
Classification example



$$z = w_1 x_1 + w_2 x_2 + b$$

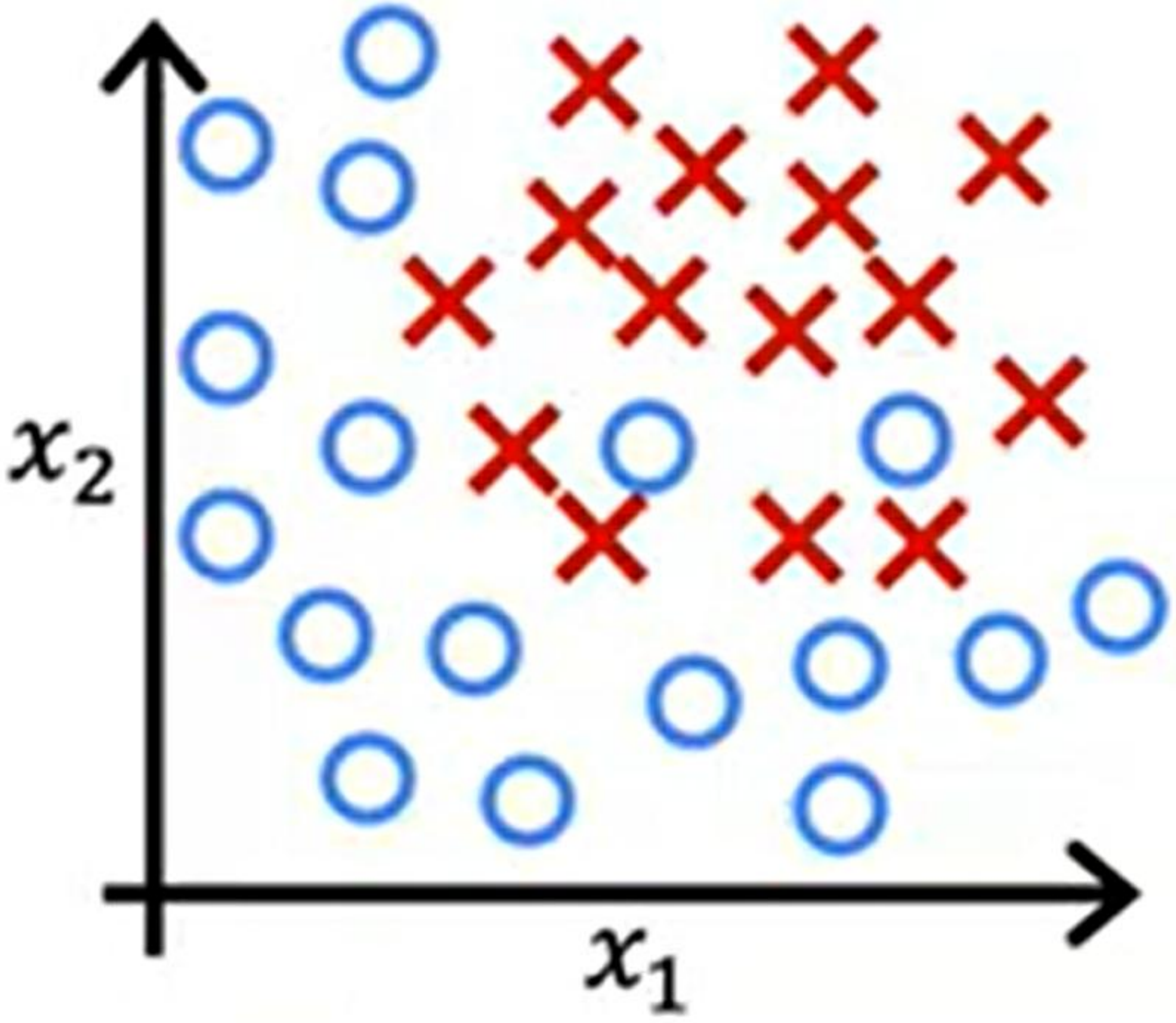
$$f_{\vec{w},b}(\vec{x}) = g(z)$$

g is the sigmoid function

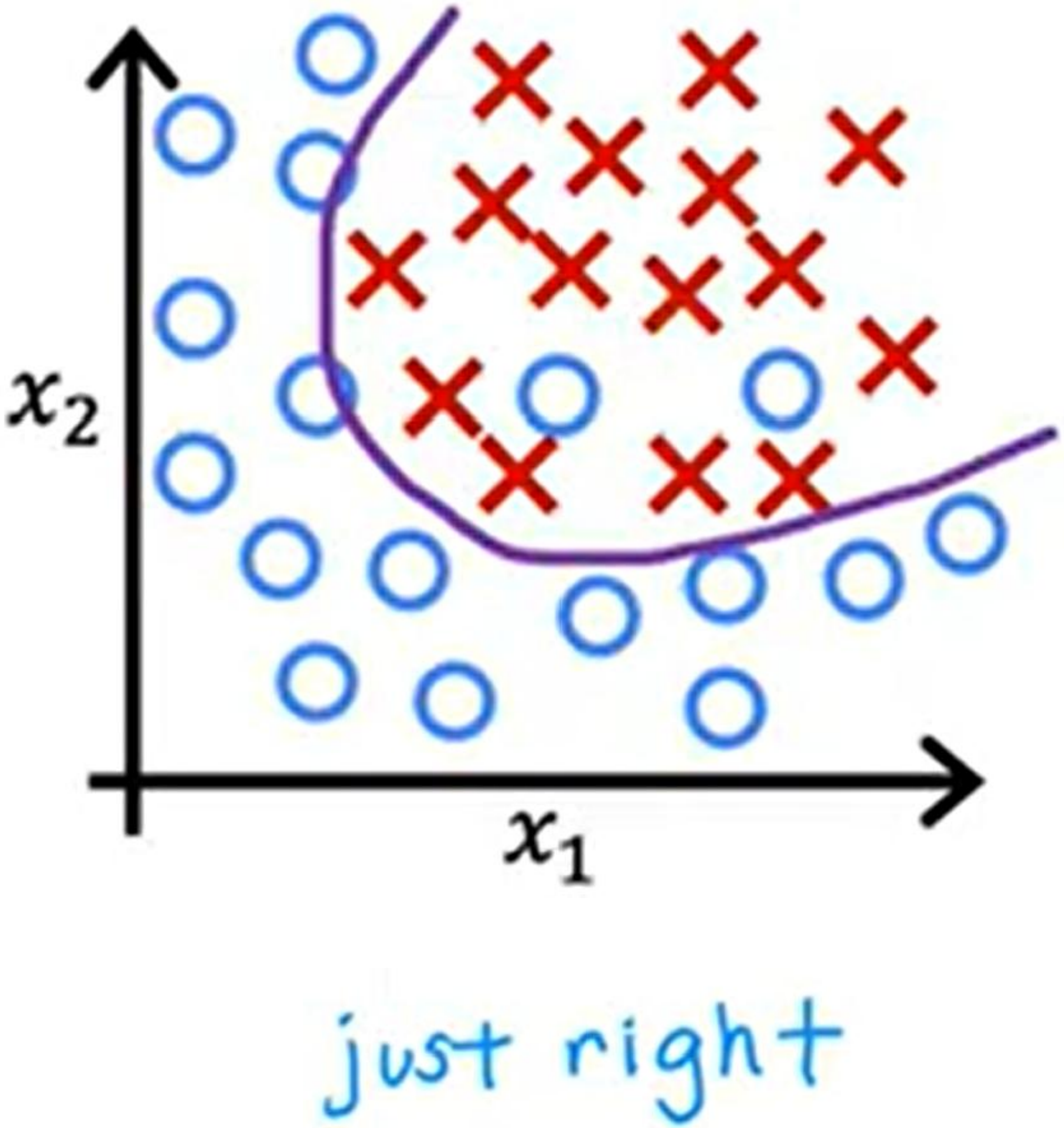


underfit high bias

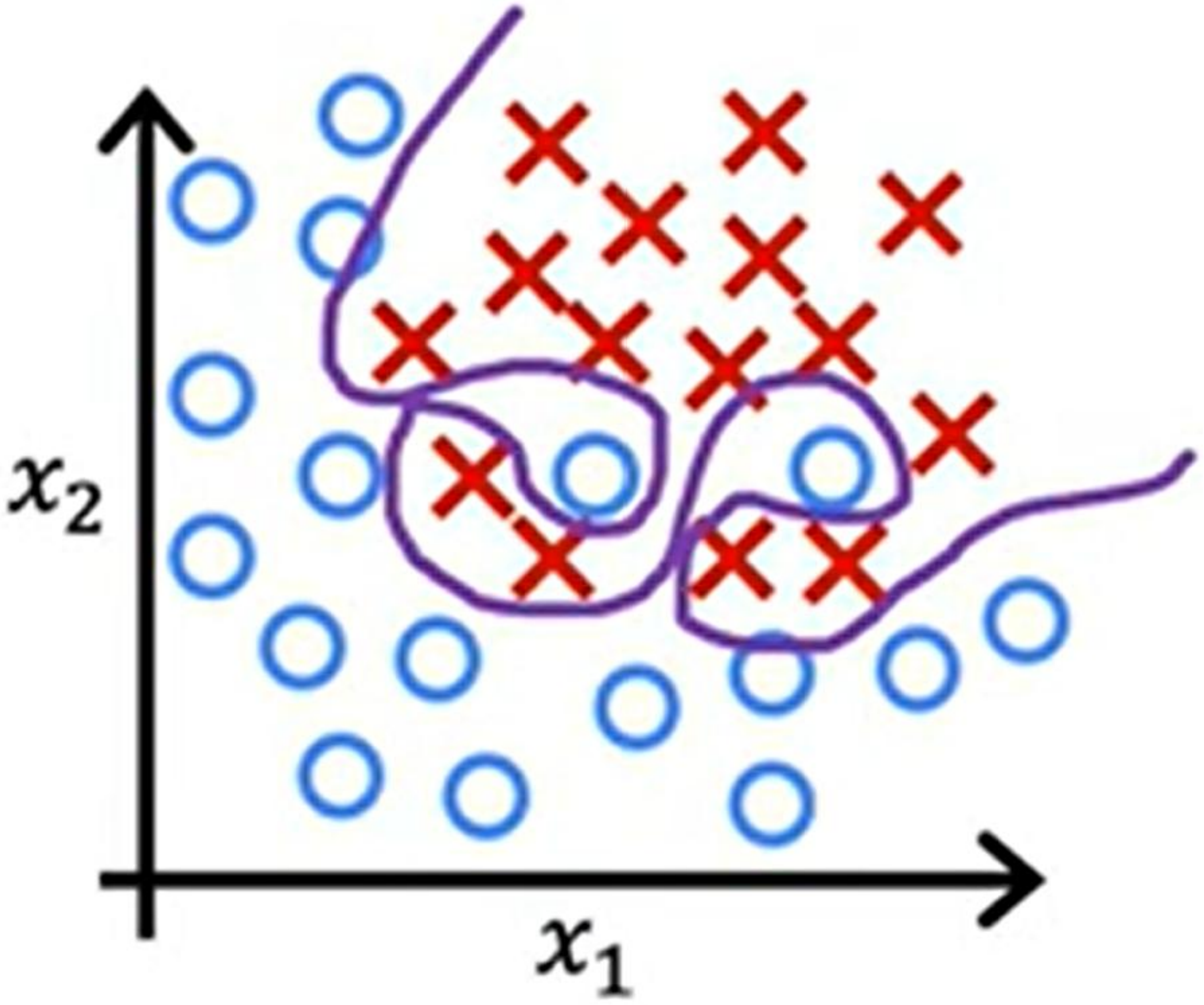
Classification example



$$z = w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2 + b$$



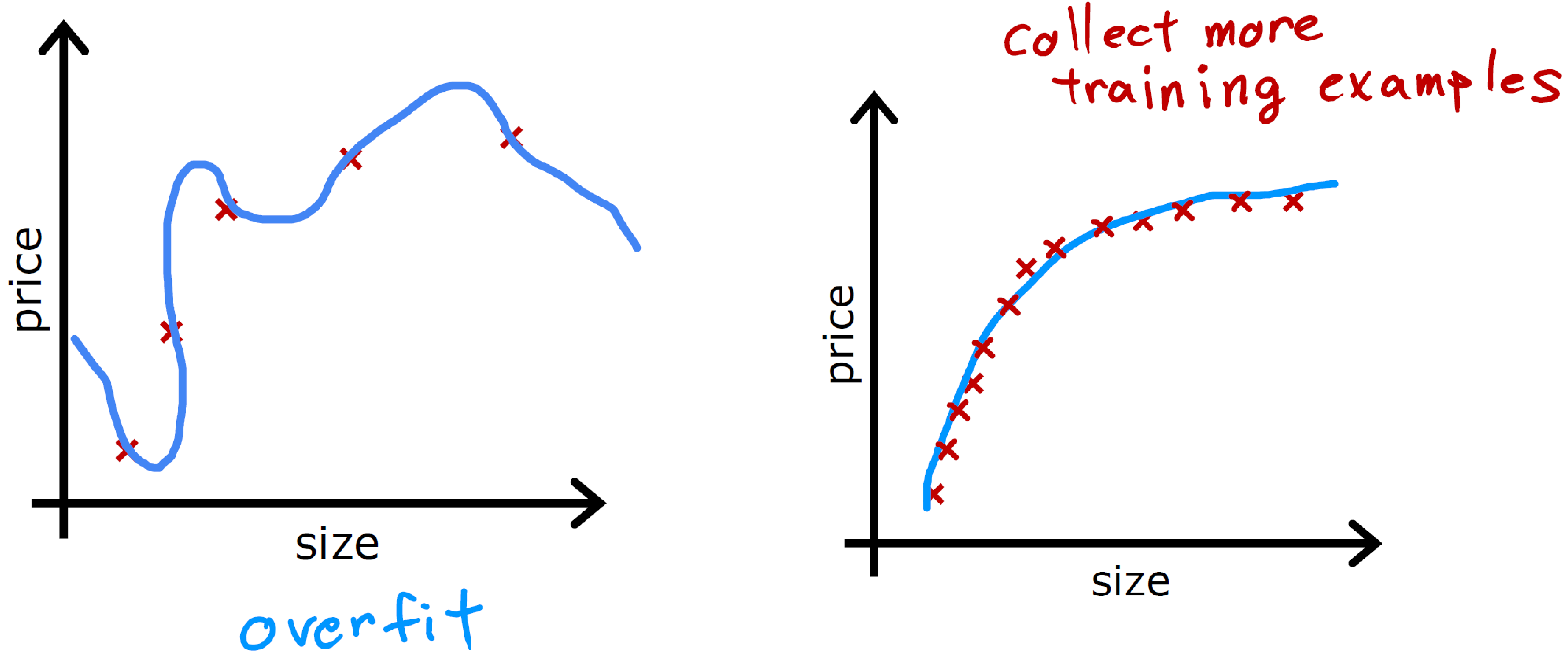
Classification example



$$
\begin{aligned}
z = {} & w_1x_1 + w_2x_2 \\
& + w_3x_1^2x_2 + w_4x_1^2x_2^2 \\
& + w_5x_1^2x_2^3 + w_6x_1^3x_2 \\
& + \cdots + b
\end{aligned}
$$

Overfit

Addressing Overfitting



Addressing Overfitting

Select features to include/exclude

size	bedrooms	floors	age	avg income	…	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

all features

+

insufficient data

↓

overfit

selected features

size
bedrooms
age
just right
feature selection

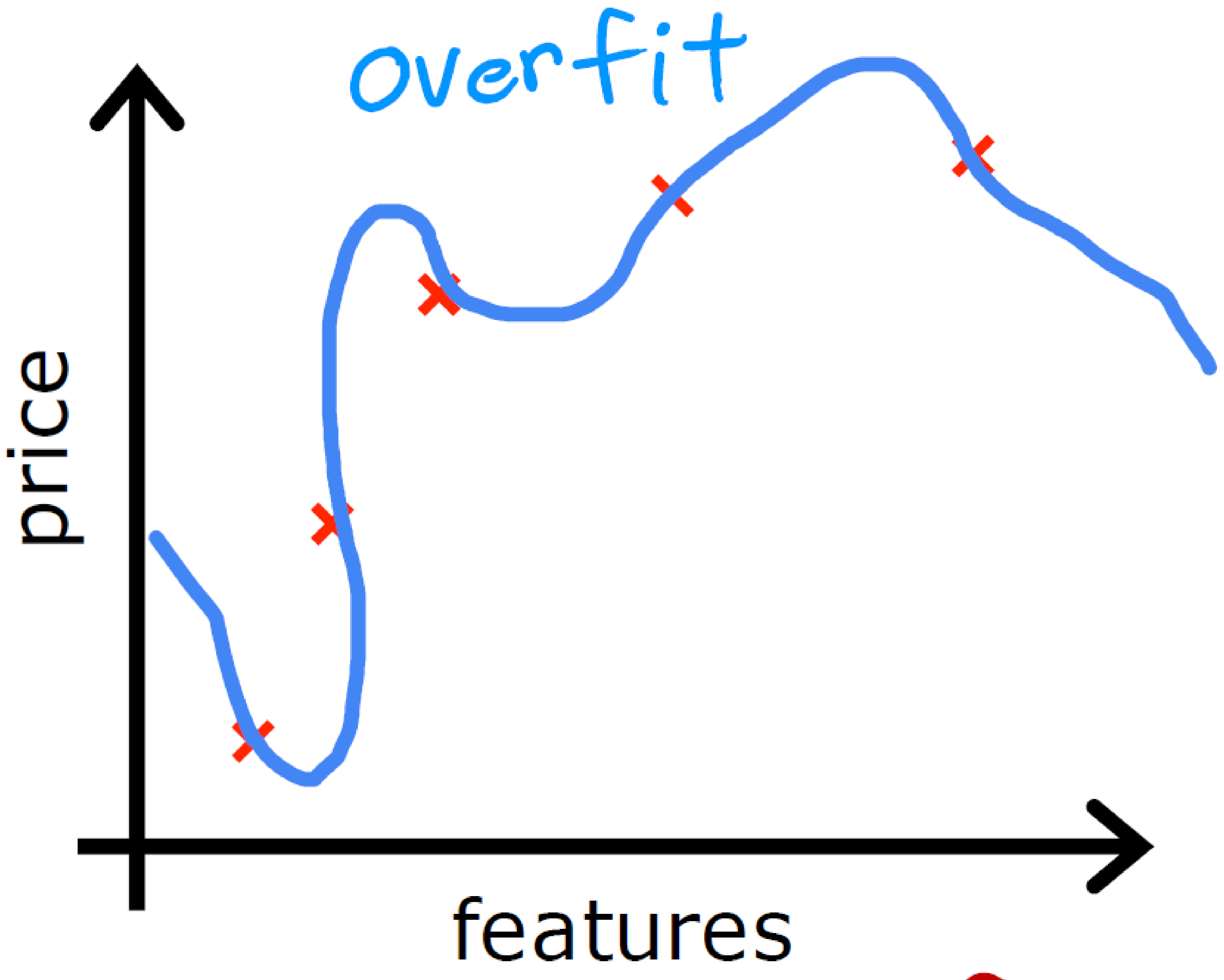
disadvantage

↓

useful features could be lost

Regularization

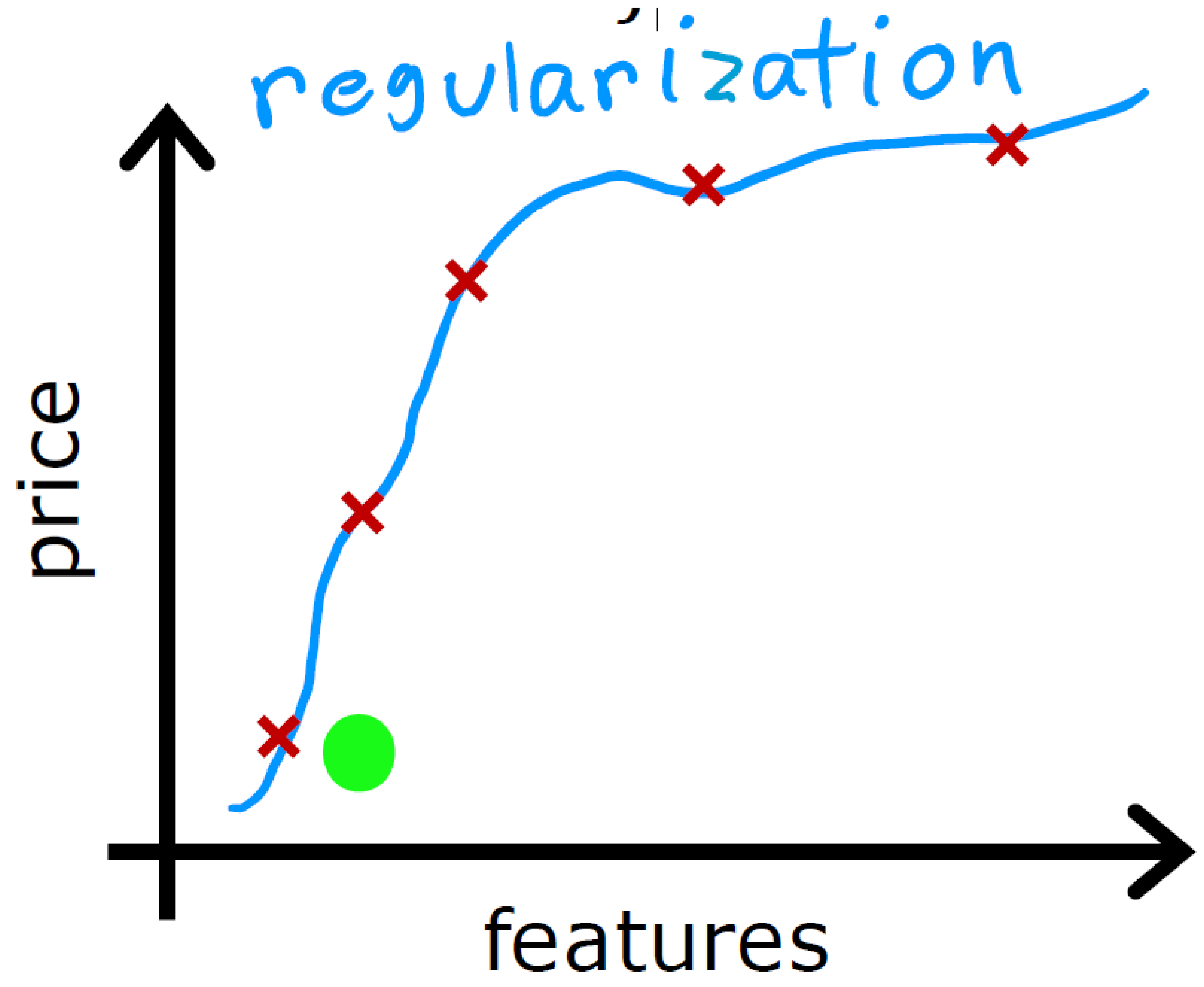
Reduce the size of parameters w_j



$$f(x) = 28x - 385x^2 + 39x^3 - \underbrace{\cancel{174}x^4}_{0} + 100$$

eliminate feature

large values for w_j



$$f(x) = 13x - 0.23x^2 + \underbrace{0.000014}x^3 - \underbrace{0.0001}x^4 + 10$$

small values for w_j

Addressing Overfitting

Options

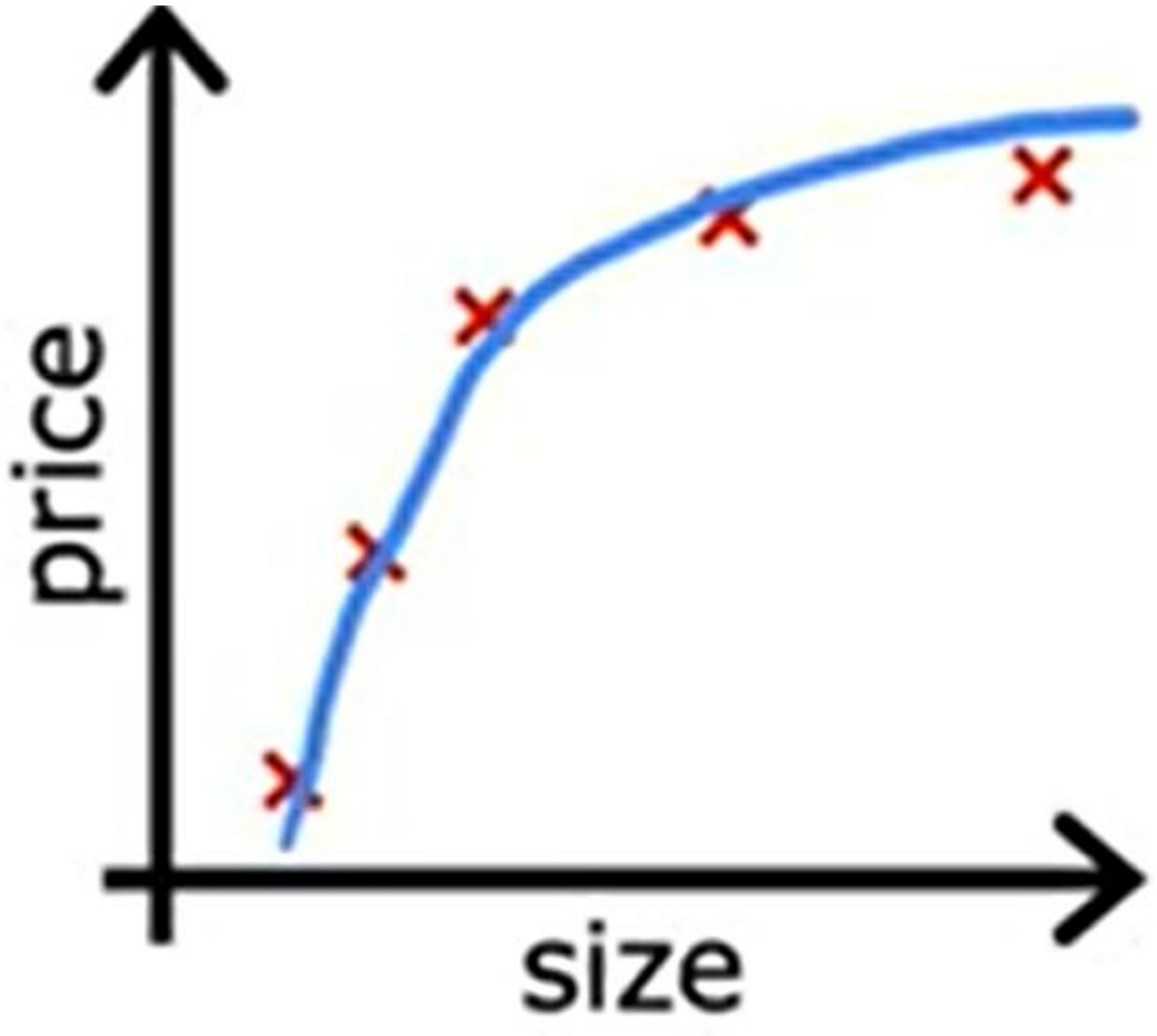
1. Collect more data

2. Select features
   – Feature selection

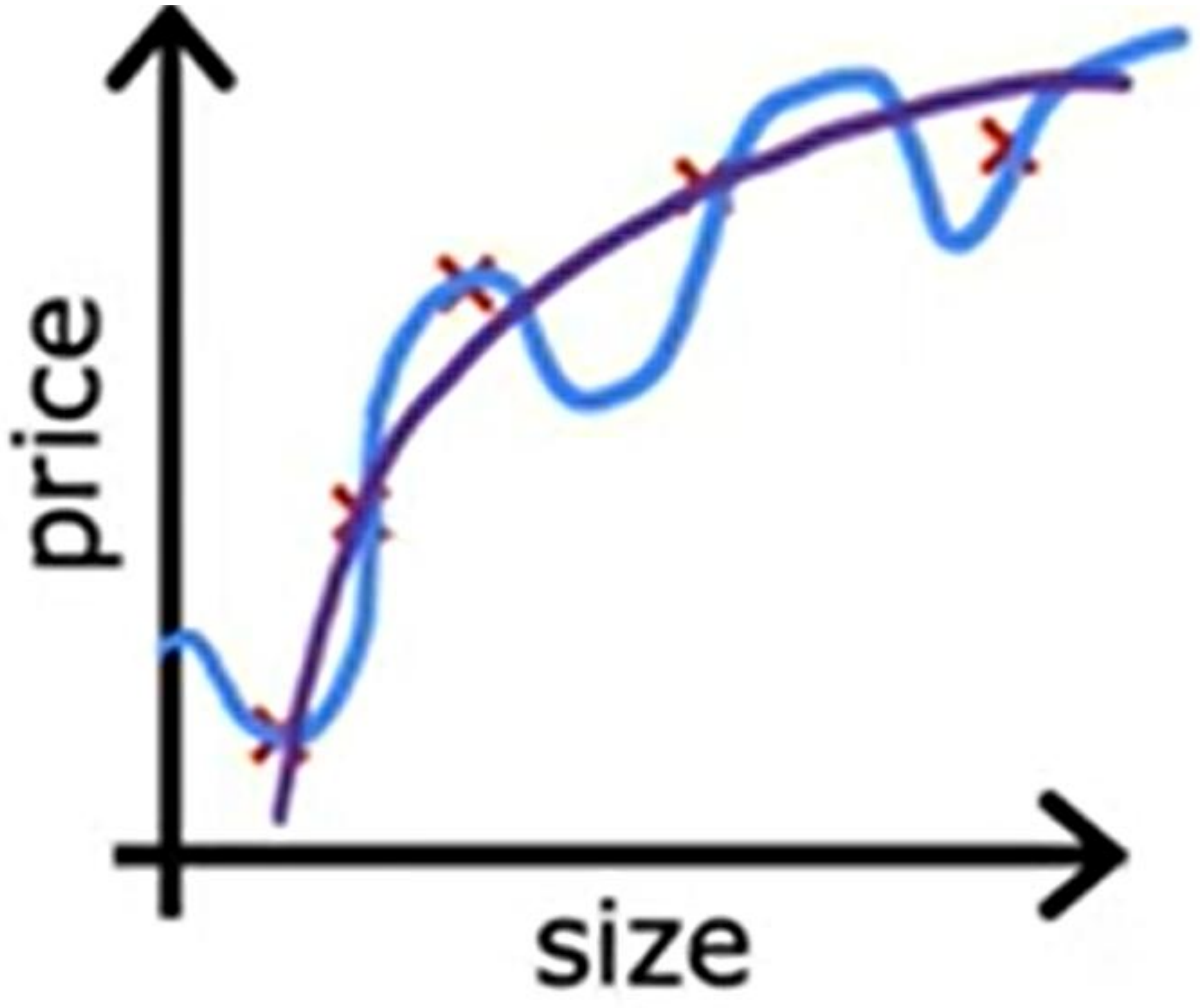
3. Reduce size of parameters

   – “Regularization”

Cost Function with Regularization



$$w_1x + w_2x^2 + b$$



$$w_1x + w_2x^2 + \underbrace{\cancel{w_3x^3}}_{\approx 0} + \underbrace{\cancel{w_4x^4}}_{\approx 0} + b$$

make w_3, w_4 really small (≈ 0)

$$\min_{\vec{w},b} \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + 1000\underbrace{w_3^2}_{0.001} + 1000\underbrace{w_4^2}_{0.002}$$

Cost Function with Regularization

small values $w_1, w_2, \cdots, w_n, b$ less likely to overfit

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

$w_1, w_1, w_2, \cdots, w_{100}, b$

n features

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \overbrace{\underbrace{\frac{\lambda}{2m}}_{\text{"lambda"}}\sum_{j=1}^{n} w_j^2}^{\text{regularization term}} + \frac{\lambda}{2m} b^2$$

can include or exclude b

"lambda" regularization parameter $\lambda > 0$

Cost Function with Regularization

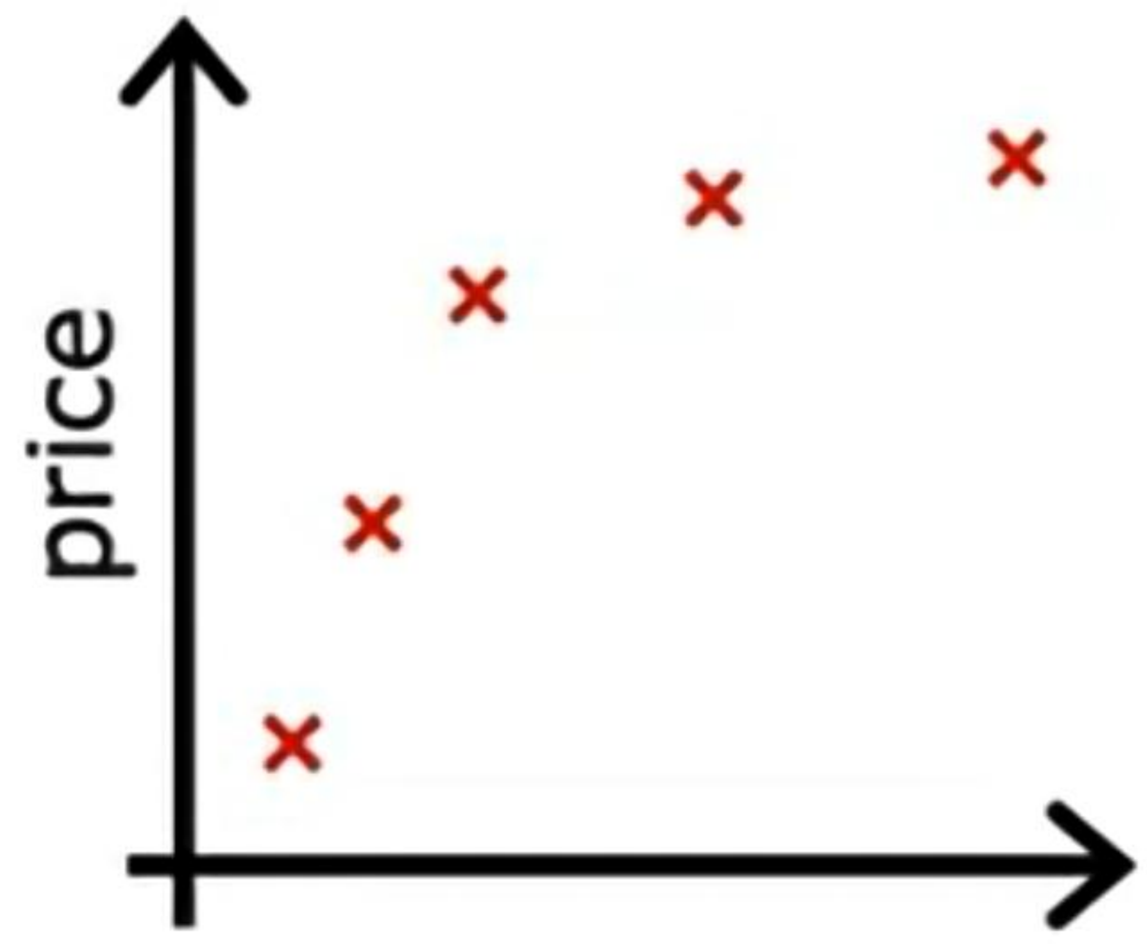
Regularization

$$\min_{\vec{w},b} J(\vec{w},b) = \min_{\vec{w},b} \left[\overbrace{\frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2}^{\text{mean squared error}} + \overbrace{\frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2}^{\text{regularization term}} \right]$$

fit data

Keep w_j small

λ balances both goals



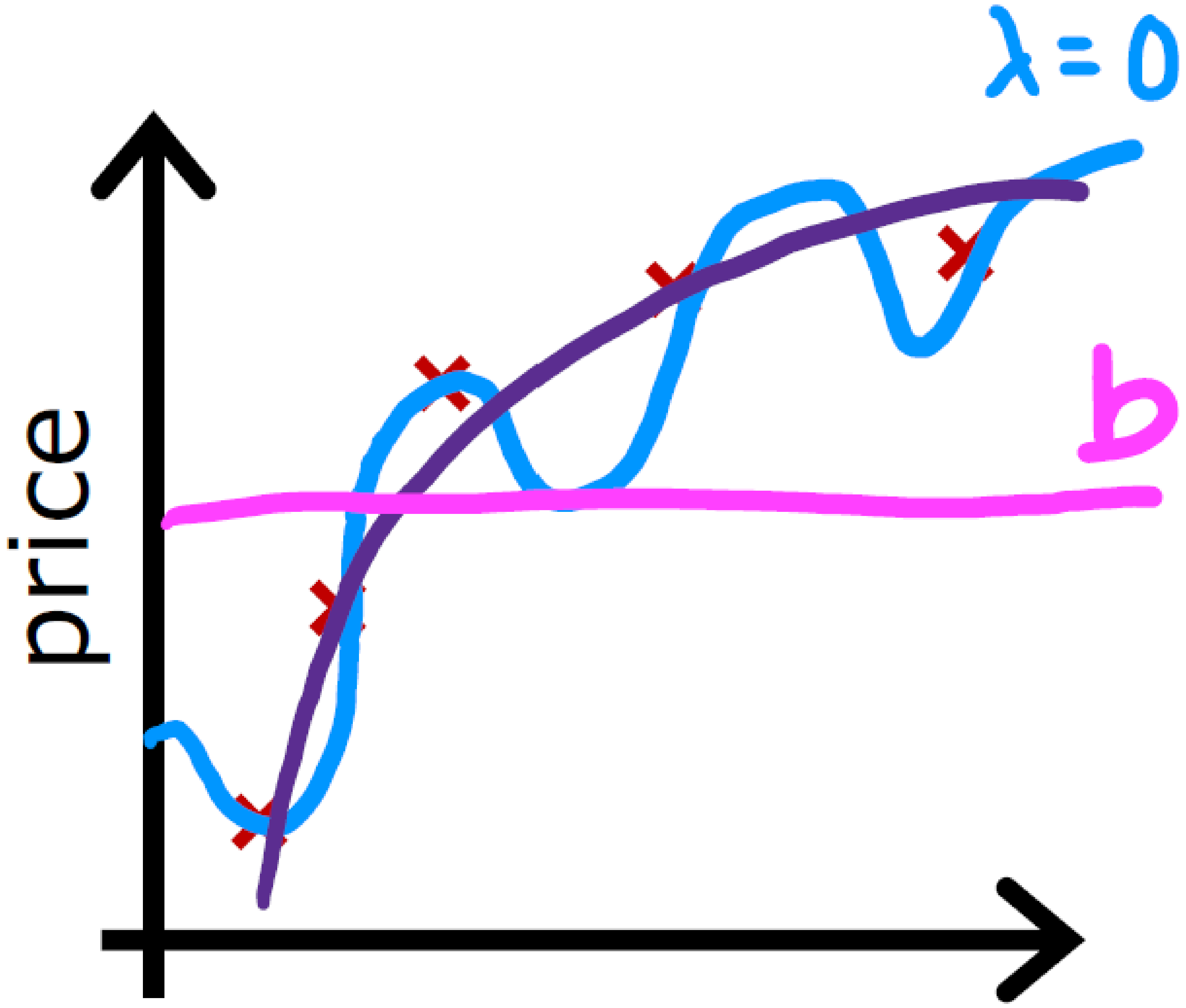
$$f_{\vec{w},b}(\vec{x}) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$$

Cost Function with Regularization

choose $\lambda = 10^{10}$

$$f_{\vec{w},b}(\vec{x}) = \underbrace{\cancel{w_1}}_{\approx 0} x + \underbrace{\cancel{w_2}}_{\approx 0} x^2 + \underbrace{\cancel{w_3}}_{\approx 0} x^3 + \underbrace{\cancel{w_4}}_{\approx 0} x^4 + b$$

Cost Function with Regularization



Choose λ

Regularized Linear Regression

$$\min_{\vec{w},b} J(\vec{w},b) = \min_{\vec{w},b}\left[\frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)})-y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2\right]$$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w},b) \qquad = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)})-y^{(i)}\right)x_j^{(i)} + \frac{\lambda}{m} w_j$$

$j=1,\ldots,n$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w},b) \qquad = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)})-y^{(i)}\right)$$

don't have to regularize b

} simultaneous update

Implementing gradient descent

repeat {

- $$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} \left[\left(f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right) - y^{(i)} \right) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right) - y^{(i)} \right)$$

} simultaneous update $j = 1 \ldots n$

Implementing gradient descent

repeat {

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} \left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)$$

} simultaneous update $j = 1 \ldots n$

$$w_j = \underbrace{1 w_j - \alpha \frac{\lambda}{m} w_j}_{w_j \underbrace{\left(1 - \alpha \frac{\lambda}{m}\right)}_{\text{shrink } w_j}} \underbrace{- \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}}_{\text{usual update}}$$

$$\alpha \frac{\lambda}{m}$$

$$0.01 \frac{1}{50} = 0.0002$$

$$w_j \left(\underbrace{1 - 0.0002}_{0.9998} \right)$$

How we get the derivative term

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{\partial}{\partial w_j}\left[\frac{1}{2m}\sum_{i=1}^{m}\Big(\underbrace{f(\vec{x}^{(i)})}_{\vec{w}\cdot\vec{x}^{(i)}+b} - y^{(i)}\Big)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2\right]$$

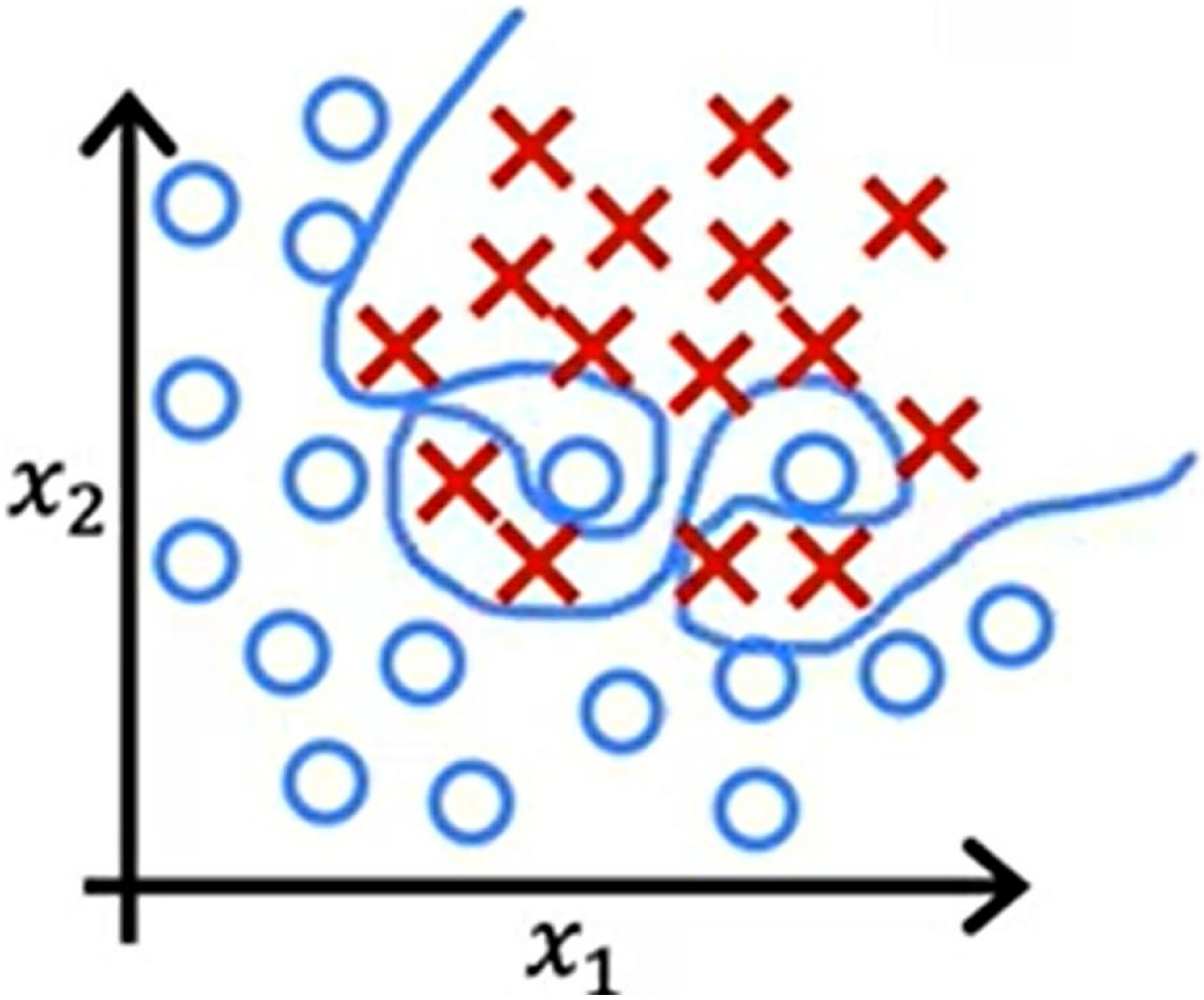
$$= \frac{1}{\cancel{2}m}\sum_{i=1}^{m}\left[\left(\vec{w}\cdot\vec{x}^{(i)} + b - y^{(i)}\right)\cancel{2}\, x_j^{(i)}\right] + \frac{\lambda}{\cancel{2}m}\cancel{2}\, w_j$$

No $\sum_{j=1}^{n}$

$$= \frac{1}{m}\sum_{i=1}^{m}\left[\Big(\underbrace{\vec{w}\cdot\vec{x}^{(i)} + b}_{f(\vec{x})} - y^{(i)}\Big) x_j^{(i)}\right] + \frac{\lambda}{m} w_j$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}\right] + \frac{\lambda}{m} w_j$$

Regularized logistic regression

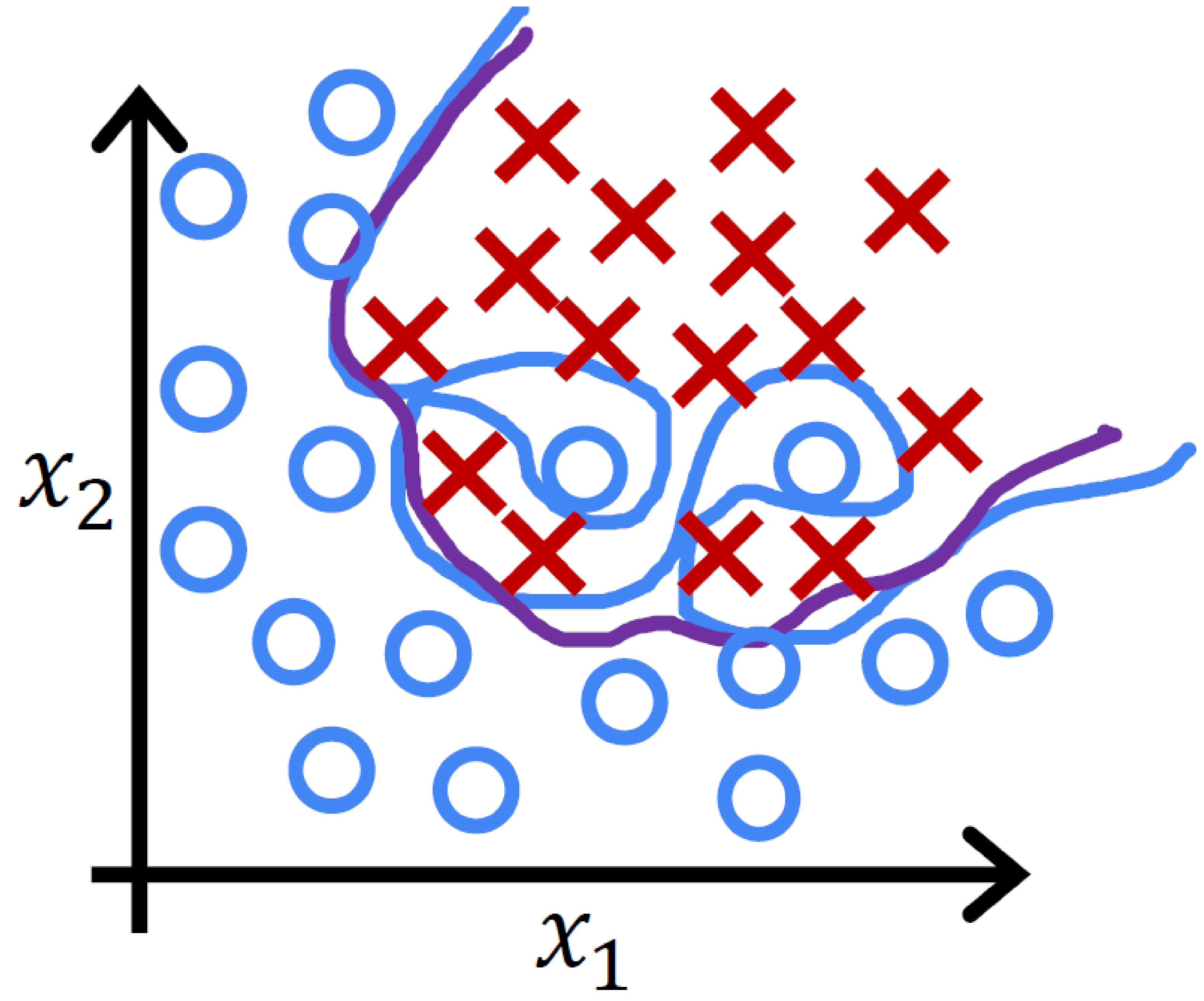


$$z = w_1x_1 + w_2x_2 + w_3x_1^2x_2 + w_4x_1^2x_2^2 + w_5x_1^2x_2^3 + \cdots + b$$

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1+e^{-z}}$$

$$J(\vec{w},b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1-f_{\vec{w},b}(\vec{x}^{(i)})\right)\right]$$

Regularized logistic regression



$$z = w_1x_1 + w_2x_2 + w_3x_1^2x_2 + w_4x_1^2x_2^2 + w_5x_1^2x_2^3 + \cdots + b$$

$$f_{\vec{\mathrm{w}},b}(\vec{\mathrm{x}}) = \frac{1}{1+e^{-z}}$$

Cost function

$$J(\vec{\mathrm{w}},b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right)\right) + \left(1-y^{(i)}\right)\log\left(1-f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right)\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

$$\min_{\vec{w},b} J(\vec{\mathrm{w}},b) \rightarrow w_j \downarrow$$

Regularized logistic regression

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1 - f_{\vec{w},b}(\vec{x}^{(i)})\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

$\min_{\vec{w},b}$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$j = 1 \ldots n$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

}

Looks same as for linear regression!

$$= \frac{1}{m}\sum_{i=1}^{m}\left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}\right] + \frac{\lambda}{m} w_j$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

logistic regression

don't have to regularize b

Thank You

Any questions?

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Source & Special Thanks to Andrew Ng (Coursera)
Machine Learning Course
https://www.youtube.com/@Deeplearningai